TraceRoute: Testing Work-Aware Reviewer Routing

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With Apart Research

Development study

1 Introduction

Can the organization of distributed work improve the verification of an agent's next step, rather than merely add another critic or more model calls? TraceRoute studies a narrow mechanism: different working branches acquire different results, constraints, counterexamples, and observations. A reviewer selected using its actual work profile may consequently possess a better basis for checking a proposal than a randomly selected member of the same pool.

The proposed chain is distributed work → differentiated evidence → reviewer routing → evidence use → defect detection. The primary comparison holds the realized worker pool, model, reviewer count, review instructions, and budget caps fixed. Fresh-context review is diagnostic. We do not test swarm superiority over a direct solver, nor whether knowledge must remain with the agent that originally acquired it. Detection and preventing an erroneous action are separate endpoints.

This report separates implementation validation, scientific grading of previously saved submissions, and the new routing experiment. A successful model call, valid citation, executable artifact, or high benchmark score cannot substitute for a measured routing effect. All five selected scientific tasks have already been used during development; their results are not independent confirmation.

2 Related Work

Greenblatt et al. [3] evaluate control protocols by harmful outcomes and retained usefulness, rather than persuasive criticism alone. Their unsuccessful producer-justification variant motivates requiring checkable grounds for objections. Foerster et al. [6] distinguish architectural control-flow protection from correctness of the data selecting an allowed branch. We transfer this distinction to host-enforced reference integrity versus fallible semantic judgments, not to a claim that provenance guarantees truth.

Rose et al. [2] explicitly distinguish causal collusion from an operational communication-and-harm criterion and show sensitivity to the capabilities and instructions of transcript judges. Zhang et al. [4] study failure localization using expert annotations; agreement with an attributed decisive step is not an intervention demonstrating that changing it repairs the task. Together these motivate separating observable links of the mechanism and retaining uncertain judgments.

Riedl [5] finds changes in coordination without a significant overall success-rate difference between main treatments. We therefore use realized work and evidence, not persona diversity or synergy as a proxy for verification success. Motwani et al. [1] motivate attention to shared context and information asymmetry, but do not establish collusion in our cooperative setting. Mediated MARL [7] and Stackelberg attacks [8] concern incentives, authority, and commitments under different assumptions; they inform future adversarial evaluations rather than the present routing baseline. None of these eight works directly evaluates our matched-pool contrast.

*Research conducted at the Swarm Alignment Hackathon, November 2026

3 Methodology

3.1 Tasks, models, and sampling

The five episodes, fixed before new review outcomes, are SCI-AGENT-EXAM [9] bioinformatics error diagnosis, chemistry result interpretation, materials-science literature synthesis, mathematics code generation, and physics-simulation tool orchestration. Selection retains the prior five-task order and does not depend on which tasks previously favored routing. No failed episode is replaced. All five preassign the `unmet_precondition` defect type; the experiment does not cover the full defect taxonomy.

Each episode assigns eight real working branches: baseline, provenance, prerequisites, alternative explanation, measurement, robustness, resources, and decision link. Workers, author, and reviewers use `openai-codex/gpt-6.1-terra`. Independent fresh contexts of `openai-codex/gpt-6-astra` validate cards and provide primary evaluation; `openai-codex/gpt-6.1-sol` supplies cross-model calibration and adjudication. These are exact subscription selectors, not assumed equivalent model families.

Worker branches execute once in isolated offline Docker environments. Their observed files, tool results, evidence, and completed states are frozen. The author proposes a next step; required profile tags are derived from the accepted valid proposal before defect injection. The router receives profiles and requirements, not hidden defect labels. All task pools, cards, and panel assignments are frozen before the first experimental review.

3.2 Conditions and budgets

`RandomHistory` samples two distinct eligible workers without replacement. `WorkProfileOverlapRouted` selects maximum tag overlap, then complementary uncovered overlap, with frozen tie-breaking and fallback. Its stored implementation name is `RoutedHistory`. `Fresh` receives no worker state. Each task has valid/defective variants, three conditions, two repeats, and two reviewers: 60 panels and 120 requested review slots. Reviews are independent before observing peer verdicts.

Workers have four turns and 8,192 output tokens per request; author and review output caps are 4,096 and 2,048. Worker and author input caps are 262,144 canonical UTF-8 bytes. Validator and evaluator output caps are 8,192 and 2,048. Full prompts, seeds, tool policies, model configuration, and source hashes are in the freeze manifest. Equal caps do not imply equal realized input length or compute; actual usage is reported separately. No automatic model retries or replacement samples are allowed.

3.3 Evidence and evaluation

Models select short local references such as `[[ev:1]]`; the host resolves them to immutable evidence and provenance within the reviewer's permitted scope. Invalid or inaccessible references are technical missingness, not scientific non-detection. Exact source binding is distinct from semantic support.

Before reviews, five frozen controls test a supported prerequisite objection, a correct proposal, an unsupported accusation, a cosmetic observation, and explicit uncertainty. Only semantically determinate axes gate acceptance; all returned axes are retained. The evaluator must pass every preregistered applicable axis. Cross-model agreement is not human validation.

Metrics distinguish informed coverage, resolved target citations, evaluator-supported target use, correct defect detection, false positives, missingness, and costs. "Informed" means registered ownership of at least one target fragment; private coverage excludes evidence judged visible in the public card context. A task, not a review, is the paired unit. Random coverage is $1 - \binom{n-k}{2}/\binom{n}{2}$ for k informed members of a pool of n. Detection requires a supported, decision-relevant, non-uncertain

objection of an accepted target type. Positive claims on valid cards require adjudication. Where paired data exist, 95% task-bootstrap intervals use 2,000 resamples and seed 20261108; missing pairs are not zeros, and small-sample intervals are descriptive.

The development transition gate requires improved routed coverage on at least four of five tasks, mean paired detection gain ≥ 0.25, routed false-positive rate ≤ 0.10 and no greater than random, differentiated evidence, and complete usable measurement. A separately preregistered evidence-removal diagnostic selects the first eligible task/owner without using detection outcomes; residual equivalent evidence or an unavailable matched sham limits causal interpretation.

4 Results

4.1 Completed execution, but no positive routing PoM

All 40 worker assignments were attempted once; 33 completed. Seven remained missing: three closed-sandbox failures, two tool-output budget overflows, one non-matching source fragment, and one unobserved evidence source. Two append-only pre-review amendments classified proven terminal local/contract failures without retrying workers or increasing budgets. Earlier artifacts remained hash-identical. All five author/card preparations succeeded, all 60 panels were frozen, and all 120 requested reviews completed with accepted evidence-reference contracts.

The decisive mechanism failure occurred before reviewer performance: all target evidence was judged visible in the public card context. Every task therefore had zero private-informed workers and zero routed private-coverage gain. Registered ownership still differed, but reviewers did not need those owners to obtain the target grounds. The predefined private-evidence intervention had no eligible case and dispatched no reviews. This is unavailability, not a measured zero causal effect.

Task	Workers / 8	Citation Δ	Support-proxy Δ
Bioinformatics	6	+0.50	+0.25
Chemistry	6	−0.25	−0.25
Materials science	6	0	0
Mathematics	7	−0.25	−0.25
Physics simulation	8	0	0

Deltas are Routed minus Random on defective-card reviews, averaged within each task. Target-citation rates were 0.20 in both history conditions; the paired-difference sample SD was 0.306. The supported-use proxy was 0.10 for Random and 0.05 for Routed: mean difference −0.05, paired SD 0.209 across five tasks. These are descriptive public-evidence diagnostics, not evidence that routing caused better private-ground use.

4.2 Detection, baselines, and measurement uncertainty

Condition	Defect panels detected / scored	Unscored	Requested
Fresh	10/10	0	10
RandomHistory	8/8	2	10
RoutedHistory	7/7	3	10

Fresh already reached the observed panel-detection ceiling. Only chemistry and mathematics had both repeats scored in both history conditions. Their paired detection differences were zero; the two-task bootstrap interval is $[0, 0]$, a degenerate selected-subset diagnostic, not evidence of population equivalence. The five-task primary contrast remains unmeasured.

There were 120 primary labels, 24 retests, two 36-item cross-model calibration lanes, and 63 adjudications. Nine adjudications remained uncertain, leaving 111 final-scored reviews and eight panels without complete final measurement. The frozen global `fp_report_ready` rule requires all mandatory adjudications to resolve; consequently all primary false-positive rates remain unavailable. We did not relax this rule after observing results or equate missing values with zero.

All five initial evaluator controls passed. Retest agreement was 24/24 for significance, type, and decision relevance, but 21/24 for evidence support. Cross-model support agreement was 31/36 and 35/36 in the two lanes, despite perfect significance/type agreement. Thus basic calibration and reliable transport did not eliminate scientific uncertainty. The positive gate failed; no downstream science action or action-gate experiment was executed.

4.3 Actual costs and auxiliary checks

Review condition	Calls	Input tokens	Output tokens	Call-seconds
Fresh	40	132,867	7,729	265.67
RandomHistory	40	547,957	8,144	240.82
RoutedHistory	40	511,645	8,047	250.67

Across preparation, reviews, and measurement, the new run used 534 calls, 3,473,400 input tokens, and 150,935 output tokens, with no missing usage records. Summed provider latency was 6,847.01 seconds, not wall time. The three execution commands took 6,646.00 seconds in total, excluding operator pauses. Five earlier calibration calls are shared and counted separately. Equal review caps did not produce identical realized token use.

The integrated suite passed 637 tests and a later transport-focused run passed 99. Real smoke verification preserved scientific `E1` while resolving `[[ev:1]]`. Separate Astra grading of five saved solver submissions yielded macro weighted score 0.6661; Appendix B reports those scores and packet-visibility limitations. They are not a routing baseline.

5 Discussion

The experiment did not establish the proposed routing advantage. It also did not isolate a meaningful test of private-work evidence delivery: the constructed public cards exposed the relevant grounds in every episode. In bioinformatics, for example, the public card already stated the import failure while the injected precondition explicitly allowed acceptance without resolving it. A fresh reviewer could recognize the contradiction without the original worker's history. The absence of private coverage is a property of this construction, not evidence that distributed work can never add verification capability.

There are two distinct outcomes. Engineering reliability improved: all requested reviews completed with host-resolved references, and earlier samples survived continuation unchanged. Scientifically, the addressable-evidence prerequisite failed, Fresh had no observed detection headroom, and unresolved adjudication prevented the complete primary comparison. Neither the successful transport nor conditional perfect detection rates should be presented as a successful PoM.

The next useful development step is an outcome-blind eligibility checkpoint after freezing real work and constructing candidate cards, but before spending experimental reviews. It should verify that relevant grounds are genuinely differentiated, not already supplied by the common proposal, and addressable by the frozen profile vocabulary. Reject unsuitable episodes with recorded reasons; do not repeatedly regenerate cards until a preferred routing outcome appears. A subsequent confirmatory set must exclude these development-consumed episodes. Full end-to-end evaluator fixtures should also exercise unresolved adjudication and global false-positive readiness, not only individual judgment axes.

Future improvements must be chosen by the failed link. No coverage gain with genuinely private evidence would motivate routing changes. Better delivery without supported use would motivate review/measurement changes. More detection with excessive false positives would motivate calibration rather than claiming unqualified success. The present run primarily motivates fixing episode construction and measurement interpretability, not expanding into a swarm-versus-solver benchmark.

The action-gate stage remains conditional and was not entered. After a positive PoM, approval should bind to a specific artifact hash and action arguments. Reviewers recommend; a separate mechanism executes, delays, or rejects. Measure erroneous actions performed, errors prevented, correct actions blocked, useful completion, and costs. Saying "repair this" is not prevention; blocking everything is not useful safety.

Evidence ablations require removal of duplicates and derived statements, not merely reference strings. The present predeclared intervention was unavailable because no private target survived. This is not an owner-memory result. A later identifiable intervention could support a causal use claim without asserting that only the original worker can possess the knowledge.

6 Limitations and Dual-Use / Ethical Considerations

Scope and contamination. The five SCI tasks are development-consumed, heterogeneous, and small in number. New worker states or random seeds do not restore independent confirmation. Within-task repeats share tasks and working states. Descriptive paired intervals cannot justify broad significance or deployment claims. A later confirmatory study needs new, preregistered episodes and independent scientific validation.

Measurement uncertainty. Astra and Sol are fallible and may share errors. Fresh contexts do not guarantee independent reasoning; using Astra for both card validation and primary evaluation can induce correlated biases. The five objective controls are narrow. Cross-model agreement is not a human expert oracle, and a validator's target-evidence map is not itself a causal intervention. Programmatic checks and rubric grading cover different properties. Upstream packet truncation can produce evaluator omissions even when execution is correct.

Compute and mechanism. The matched routing comparison holds model, reviewer count, and budget caps constant, not every realized input token. Histories naturally differ. An observed advantage would not by itself isolate an effect beyond input length, prove swarm efficiency over a direct solver, establish superiority of a richer dependency router, or show that the original evidence owner is uniquely necessary. Ownership does not enumerate all related knowledge in a history. The semantic-use proxy combines a target citation with review-level support and relevance; it is not per-fragment causal attribution.

Provenance is not truth. A verified reference establishes which observed bytes were cited and who could access them. An agent can still create a misleading artifact, misinterpret a valid source, or cite genuine but irrelevant material. The host reference mechanism is not a defense against all prompt injection, strategic collusion, or coordinated fabrication.

Dual use. Work-aware routing could make harmful multi-agent operations more efficient by finding agents with relevant capabilities or complementary information. Manipulated profiles could direct scrutiny toward cooperative or compromised reviewers. Shared private context could enable coordinated misleading explanations while making a panel appear independent; a source-backed audit trail could create unjustified confidence and obscure semantic deception. We do not test resistance to such strategies, and no collusion-detection capability is claimed.

Operational boundaries. Experiments use offline, non-root sandboxes and public benchmark inputs; no real laboratory or external deployment action is authorized by the review outputs. Credentials and OAuth material must not enter research artifacts. Reproducibility materials should publish scientific inputs, prompts, hashes, and sanitized judgments, not account access. Any future action-gate experiment must preserve an independent execution boundary and evaluate false blocking alongside prevented harm.

What is established. The completed run demonstrates reliable review-reference transport under this setup, but fails its addressable-private-evidence prerequisite and leaves the full detection/false-positive comparison inconclusive. It establishes neither a positive routing effect nor its general impossibility, robust adversarial safety, or prevention of erroneous downstream actions.

References

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[2] A. Rose et al. (2026). *Detecting Multi-Agent Collusion Through Multi-Agent Interpretability*. arXiv:2604.01151v5. https://arxiv.org/abs/2604.01151v5.

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[6] H. Foerster et al. (2026). *CaMeLs Can Use Computers Too: System-level Security for Computer Use Agents*. arXiv:2601.09923v3. https://arxiv.org/abs/2601.09923v3.

[7] D. Ivanov, I. Zisman, and K. Chernyshev (2023). *Mediated Multi-Agent Reinforcement Learning*. AAMAS; arXiv:2306.08419v1. https://arxiv.org/abs/2306.08419v1.

[8] D. Landis and N. I. Schwartzbach (2023). *Stackelberg Attacks on Auctions and Blockchain Transaction Fee Mechanisms*. arXiv:2305.02178v1. https://arxiv.org/abs/2305.02178v1.

[9] SKYLENAGE-AI. *Sci-Agent-Exam*, public example repository. Snapshot 81f29be06ad155ce41ac4aa4f671594674ae6836, retrieved 8 November 2026. https://github.com/SKYLENAGE-AI/Sci-Agent-Exam.

A Replication and audit artifacts

All paths are repository-relative. The literature manifest records exact versions and PDF hashes in `papers/agent-collusion/manifest.json`; digitized sources are under `papers/agent-collusion/markdown/`.

The operational protocol is `docs/traceroute_routing_pom.md`. Configuration, seeds, cases, source hashes, inputs, planned slots, and budgets are in `runs/traceroute-routing-pom/execution-v2/protocol-freeze.json`; final outcomes are in that directory's `report.json` and `terminal-worker-continuation-outcome.json`. `all-task-pools-cards-panels-freeze.json` binds the common pool before reviews. Earlier controls remain under `runs/traceroute-routing-pom/execution/evaluator-controls/`, with explicit reuse provenance. Both `local-failure-continuation-freeze.json` and `terminal-worker-continuation-freeze.json` preserve pre-review amendments and prior artifact hashes. Original stop outcomes remain intact. Compact derived costs are in `runs/traceroute-routing-pom/report-data.json`.

Scientific grading is recorded in `runs/sci-astra-scientific-v2/report.json`, with per-task original packets, adapted requests, provider responses, and criterion-level judgments. Original solver outputs and official deterministic checker results remain in `runs/sci-agent-exam-executable/`. The earlier HTTP rejections are preserved in `runs/sci-astra-scientific/`.

Exact worker, author, reviewer, validator, and evaluator prompts are respectively defined in `src/little_pom/workers/__init__.py`, `src/little_pom/author/`, `src/little_pom/reviews/runner.py`, `src/little_pom/cards/__init__.py`, and `src/little_pom/measurement/__init__.py`; the protocol freeze binds their source versions. Per-call intents bind actual requests. Current source files alone are not a substitute for the frozen version when replaying a historical run.

A.1 Execution commands

These commands launch the original protocol, not permission to overwrite an existing run. A fresh reproduction must use a new directory and cannot claim the original model samples. The recorded run subsequently used `continue.py` and `continue_terminal.py` in the same runner directory; their explicitly frozen terminal-failure rules are amendments to the original launcher, not hidden retries.

```
uv run --locked --extra live python -m little_pom.scientific_judge \
  --output NEW_SCI_DIRECTORY --live
uv run --locked --extra live python \
  runs/traceroute-routing-pom/launch.py prepare \
  --run-dir NEW_POM_DIRECTORY
uv run --locked --extra live python \
  runs/traceroute-routing-pom/launch.py run \
  --run-dir NEW_POM_DIRECTORY
```

Shared prior calibration calls and newly dispatched calls are accounted separately. The source freeze and prior-artifact verification must pass before a continuation; malformed results, rejected citations, and failed workers are not silently replaced.

B Separate scientific grading of saved submissions

Five saved direct-solver submissions were graded without rerunning solvers or original isolated programmatic checks. Astra received the saved upstream scientific rubrics and original figure bytes; weighted totals use the original task-specific weights. Macro weighted score: 0.6661.

Task	Programmatic	Astra rubric	Weighted total
Bioinformatics	1.0000	0.4932	0.8226
Chemistry	0.9580	0.7590	0.8486
Materials science	0.8889	0.5512	0.6187
Mathematics	0.5253	0.3770	0.4808
Physics simulation	0.5315	0.5949	0.5600

These single samples from heterogeneous domains are not repeated estimates of one success probability. No overall binary benchmark pass threshold is defined. Only biology passed all programmatic checks. Chemistry retained a runtime disposition error; mathematics and physics failed substantial hidden behavioral checks.

The scientific model was explicitly changed from upstream `gpt-5.6-sol` to `openai-codex/gpt-6-astra`. The Codex instructions field receives an exact trusted upstream guidance prefix, with original-text reconstruction checked; submitted content remains user data. These are not official upstream-model scores or human expert judgments.

Five initial HTTP-400 dispatches produced no scores and remain recorded; their original response bodies were unavailable. The corrected explicit transport revision produced five judgments, using 136,185 input and 5,046 output tokens; summed call latency was 128.45 seconds and command wall time 129.72 seconds. No Anthropic API key or solver resampling was needed.

The adapter adds no truncation, but upstream packets already truncate biology code/data and physics time-series/trajectory material. Biology rubric criteria 8–9 explicitly fail because the repaired call site is absent from the truncated packet, whereas independent programmatic replay passed. Such failures cannot automatically be attributed to the solver. Packets were not selectively repaired and regraded after inspecting scores. This auxiliary result demonstrates why executable correctness, evaluator visibility, rubric quality, and routing benefit must remain distinct.